

Question	Answer	Mark	Mark scheme	Additional guidance
8	44.9	P1	for process to find an expression for the area of the trapezium, eg $\frac{1}{2}(12 + CD)8$ or $8 \times 12 + \frac{1}{2} \times 8 \times x$ or for process to find the area of the triangle, eg $112 - 8 \times 12 (= 16)$	x is the length of the line from C to where perpendicular from B meets CD Allow use of other letters in place of CD and x (do not have to be defined unless otherwise stated) Award P1 for $112 - 8 \times 12 (= 16)$ even if not used
		P1	for forming an equation and isolating terms in the unknown length, eg $4CD = 112 - 48$ or $\frac{1}{2} \times 8 \times x = 112 - 8 \times 12$ or $\frac{1}{2} \times 8 \times x = "16"$ or $CD = 16$ or $x = 4$	Award P2 for $CD = 16$ or $x = 4$ even if not used unless clearly from incorrect working eg $12 - 8 (= 4)$ Only award P2 for 16 if it is clearly identified as CD
		P1	for start of process to find length of BC , eg $8^2 + "4"{}^2 (= 80)$ or $8^2 + [\text{their } x]^2$ or $\tan^{-1}\left(\frac{"4"}{8}\right) (= 26.5\dots)$ oe or $\tan^{-1}\left(\frac{8}{"4"}\right) (= 63.4\dots)$ where "4" can be [their x]	[their x] can be any value less than 12 or clearly identified as the length of the line from C to where perpendicular from B meets CD (may be seen on the diagram)
		P1	for $\sqrt{8^2 + "4"{}^2}$ or $\sqrt{64 + "16"}$ or $\sqrt{80}$ or $4\sqrt{5} (= 8.9\dots)$ oe or $\sqrt{8^2 + [\text{their } x]^2}$ or $\frac{"4"}{\sin "26.5\dots"}$ or $\frac{8}{\cos "26.5\dots"}$ or $\frac{8}{\sin "63.4\dots"}$ or $\frac{"4"}{\cos "63.4\dots"}$ where "4" can be [their x]	Award P4 for $(BC =) \sqrt{80}$ or $4\sqrt{5}$ or 8.9... unless $x = 4$ is clearly from incorrect working
		A1	for answer in the range 44.9 to 44.95	If an answer is shown in the range in working and then incorrectly rounded award full marks